Supplementary Information

Anthrax-MLNet: A Hybrid Machine Learning Approach with Multi-Step Feature Selection for Identifying Candidate Biomarkers and Drug Targets in Anthrax

This document provides the complete numerical results, extended analyses, and reproducibility information supporting the main manuscript. Each section states what was evaluated, how, what was found, and what the finding means, so that it can be read independently of the main text. Analyses that did not improve performance are included rather than omitted; where an analysis cannot be scored, that is reported in place of a number.

Supplementary tables are numbered S1–S12 and supplementary figures S1–S6, in order of appearance.

Table S1: The 12-gene candidate biomarker panel. Cohen's d, p-values, and false discovery rates refer to the combined discovery cohort (n = 11); per-cohort log fold-changes are reported separately for GSE14390 and GSE12131. Because the two series were deposited at different processing stages, only the direction of effect is compared between them. WNT6 is also a member of the PPI-derived hub set.

Gene	Set	Cohens_d_combined	P_Combined	FDR_Combined	logFC_GSE14390	P_GSE14390	Direction_GSE14390	logFC_GSE12131	P_GSE12131	Direction_GSE12131	Direction_concordant
LOC101928461	Common DEG	2.251	0.004	0.03	1.991	0.003	Up in exposed	0.85	0.619	Up in exposed	Yes
LINC00485	Common DEG	-2.188	0.005	0.03	-2.01	<0.001	Down in exposed	-0.769	0.631	Down in exposed	Yes
LOC100506530	Common DEG	-2.115	0.015	0.036	-2.016	<0.001	Down in exposed	-0.699	0.658	Down in exposed	Yes
WNT6	Common DEG + hub	2.068	0.011	0.036	1.326	0.003	Up in exposed	0.05	0.871	Up in exposed	Yes
RNF128	Common DEG	2.056	0.012	0.036	1.975	<0.001	Up in exposed	0.718	0.642	Up in exposed	Yes
VWC2	Common DEG	-1.922	0.027	0.053	-1.989	<0.001	Down in exposed	-0.571	0.73	Down in exposed	Yes
GULP1	Common DEG	1.526	0.032	0.055	0.323	0.394	Up in exposed	1.054	0.002	Up in exposed	Yes

CNTN4	Common DEG	1.007	0.122	0.183	1.137	0.045	Up in exposed	-0.228	0.734	Down in exposed	No
TTC6	Common DEG	0.955	0.188	0.226	0.906	0.113	Up in exposed	-0.039	0.967	Down in exposed	No
GATAD2A	Common DEG	-0.935	0.143	0.191	-0.396	0.403	Down in exposed	-0.72	0.392	Down in exposed	Yes
IL20RA	Common DEG	-0.91	0.217	0.237	-1.216	0.194	Down in exposed	0.369	0.346	Up in exposed	No
ACSS1	Common DEG	-0.376	0.54	0.54	-0.4	0.195	Down in exposed	0.325	0.283	Up in exposed	No

Table S2: Composition and processing of the two discovery cohorts. The log2 column records whether the deposited values required transformation before standardisation.

Cohort	Samples	Exposed	Control	Probe_sets	log2_transformed
GSE14390	7	3	4	42904	FALSE
GSE12131	4	2	2	42904	TRUE

Table S3: Sample annotation for the combined discovery cohort. Sample indices correspond to those used in Table S5.

Sample_index	Dataset	Group
1	GSE14390	Control
2	GSE14390	Control
3	GSE14390	Control
4	GSE14390	Control
5	GSE14390	Exposed
6	GSE14390	Exposed
7	GSE14390	Exposed
8	GSE12131	Control
9	GSE12131	Control
10	GSE12131	Exposed
11	GSE12131	Exposed

Table S4: Effect size, significance, and achieved statistical power for the sixteen genes assessed, comprising the twelve overlapping DEGs and the five PPI-derived hub genes, with WNT6 belonging to both. The required sample size is that needed to detect the observed effect at 80% power ($\alpha = 0.05$, two-sided). No gene met either quality-control criterion.

Gene	Set	Mean_exposed	Mean_control	Cohens_d	Effect_size	Direction	t_test_P	t_test_FDR	Wilcoxon_P	Wilcoxon_FDR	Achieved_power	Adequate_power	N_per_group_for_80pct
LOC101928461	Common DEG	0.852	-0.71	2.25	large	Up in exposed	0.004	0.039	0.03	0.081	0.91	Yes	5
LINC00485	Common DEG	-0.843	0.702	-2.188	large	Down in exposed	0.005	0.039	0.009	0.046	0.894	Yes	5
WNT6	Common DEG + hub	0.465	-0.387	2.068	large	Up in exposed	0.011	0.047	0.009	0.046	0.859	Yes	5
RNF128	Common DEG	0.821	-0.684	2.056	large	Up in exposed	0.012	0.047	0.017	0.069	0.855	Yes	5
LOC100506530	Common DEG	-0.831	0.692	-2.115	large	Down in exposed	0.015	0.048	0.009	0.046	0.874	Yes	5
VWC2	Common DEG	-0.796	0.663	-1.922	large	Down in exposed	0.026	0.071	0.082	0.188	0.806	Yes	6
GULP1	Common DEG	0.322	-0.268	1.526	large	Up in exposed	0.032	0.073	0.03	0.081	0.613	No	8
CNTN4	Common DEG	0.344	-0.287	1.007	large	Up in exposed	0.122	0.244	0.126	0.251	0.318	No	17
GATAD2A	Common DEG	-0.28	0.233	-0.935	large	Down in exposed	0.143	0.254	0.329	0.439	0.282	No	19
TTC6	Common DEG	0.303	-0.252	0.955	large	Up in exposed	0.188	0.301	0.329	0.439	0.292	No	19
IL20RA	Common DEG	-0.343	0.286	-0.91	large	Down in exposed	0.217	0.316	0.177	0.316	0.27	No	20
FZD9	Hub	0.404	-0.337	0.722	medium	Up in exposed	0.298	0.398	0.247	0.395	0.187	No	32
FZD3	Hub	0.097	-0.081	0.522	medium	Up in exposed	0.432	0.531	0.429	0.527	0.121	No	59
FZD6	Hub	0.243	-0.203	0.414	small	Up in exposed	0.519	0.576	0.662	0.757	0.094	No	93
ACSS1	Common DEG	-0.072	0.06	-0.376	small	Down in exposed	0.54	0.576	0.792	0.845	0.086	No	113
DKK4	Hub	-0.08	0.067	-0.134	negligible	Down in exposed	0.83	0.83	0.931	0.931	0.054	No	879

Table S5: Leave-one-out cross-validated predictions for each sample under both panels. Probabilities are those produced by the model refitted without that sample; calls are made at the Youden point of the cross-validated predictions.

Sample_index	Dataset	Group	LOO_Probability_12gene	LOO_Probability_hub	LOO_call_12gene	LOO_call_hub	Correct_12gene	Correct_hub
1	GSE143940	Control	0.2577	0.4804	Control	Exposed	Yes	No
2	GSE143940	Control	0.2368	0.5625	Control	Exposed	Yes	No
3	GSE143940	Control	0.1825	0.3624	Control	Control	Yes	Yes
4	GSE143940	Control	0.2843	0.3242	Control	Control	Yes	Yes
5	GSE143940	Exposed	0.8235	0.4795	Exposed	Exposed	Yes	Yes

6	GSE14390	Exposed	0.7909	0.496	Exposed	Exposed	Yes	Yes
7	GSE14390	Exposed	0.8114	0.667	Exposed	Exposed	Yes	Yes
8	GSE12131	Control	0.5308	0.5175	Exposed	Exposed	No	No
9	GSE12131	Control	0.28	0.448	Control	Control	Yes	Yes
10	GSE12131	Exposed	0.3578	0.4835	Exposed	Exposed	Yes	Yes
11	GSE12131	Exposed	0.684	0.4802	Exposed	Exposed	Yes	Yes

Table S6: Discrimination by individual hub genes. AUC was computed using standardised expression directly as the classifier score, without fitting any model, and is therefore free of resubstitution bias; confidence intervals are from stratified bootstrap resampling over 2000 replicates. FDR values are corrected across the five genes.

Gene	AUC	CI_low	CI_high	Cohens_d	Direction	t_test_P	t_test_FDR	Wilcoxon_P
WNT6	0.967	0.8	1	2.068	Higher in exposed	0.011	0.056	0.009
FZD9	0.733	0.4	1	0.722	Higher in exposed	0.298	0.649	0.247
FZD3	0.667	0.267	1	0.522	Higher in exposed	0.432	0.649	0.429
FZD6	0.6	0.2	1	0.414	Higher in exposed	0.519	0.649	0.662
DKK4	0.533	0.167	0.867	-0.134	Lower in exposed	0.83	0.83	0.931

Table S7: Approximately unbiased (AU) and bootstrap probability (BP) support for each internal node of the sample dendrogram, under both harmonisation pipelines. The root node is omitted, since it is supported trivially. Nodes with AU ≥ 0.95 are marked as supported.

Pipeline	Node	AU	BP	SE_AU	P_value	Supported_AU95
P1_filtered_data	1	0.732	0.664	0.0258	0.269	No
P1_filtered_data	2	0.958	0.786	0.0075	0.042	Yes
P1_filtered_data	3	0.962	0.564	0.0064	0.038	Yes
P1_filtered_data	4	0.919	0.692	0.0119	0.081	No
P1_filtered_data	5	0.856	0.543	0.0175	0.144	No
P1_filtered_data	6	0.798	0.475	0.0217	0.202	No
P1_filtered_data	7	0.768	0.474	0.0234	0.232	No
P1_filtered_data	8	0.912	0.575	0.0124	0.088	No
P1_filtered_data	9	0.893	0.499	0.0143	0.107	No
P2_datasets_csv	1	0.932	0.79	0.011	0.068	No

P2_datasets_csv	2	0.982	0.851	0.0041	0.018	Yes
P2_datasets_csv	3	0.916	0.545	0.012	0.085	No
P2_datasets_csv	4	0.894	0.64	0.0143	0.106	No
P2_datasets_csv	5	0.959	0.765	0.0073	0.041	Yes
P2_datasets_csv	6	0.934	0.548	0.0099	0.066	No
P2_datasets_csv	7	0.795	0.441	0.0221	0.205	No
P2_datasets_csv	8	0.766	0.532	0.0234	0.234	No
P2_datasets_csv	9	0.686	0.368	0.0279	0.314	No

Table S8: Two-cluster membership from the bootstrap dendrogram, with each sample's cohort of origin and infection status. Agreement with phenotype was 0.82; agreement with cohort of origin was 0.55.

Sample	Cluster	Dataset	Group
GSM305784	1	GSE12131	Control
GSM305785	2	GSE12131	Control
GSM305786	2	GSE12131	Anthrax
GSM305787	1	GSE12131	Anthrax
GSM359332	2	GSE14390	Control
GSM359753	2	GSE14390	Control
GSM359754	2	GSE14390	Control
GSM359755	2	GSE14390	Control
GSM359758	1	GSE14390	Anthrax
GSM359759	1	GSE14390	Anthrax
GSM359760	1	GSE14390	Anthrax

Table S9: Per-node AU values under the two harmonisation pipelines, with their difference and whether each node's support status was consistent between them.

Node	AU_pipeline1	AU_pipeline2	Delta_AU	Supported_P1	Supported_P2	Agreement
1	0.7315	0.9324	0.2009	No	No	consistent
2	0.9576	0.9816	0.024	Yes	Yes	consistent
3	0.9623	0.9155	0.0468	Yes	No	FLIPPED
4	0.9191	0.8939	0.0252	No	No	consistent
5	0.8559	0.9586	0.1027	No	Yes	FLIPPED
6	0.7979	0.9344	0.1365	No	No	consistent
7	0.7683	0.7949	0.0266	No	No	consistent
8	0.9117	0.7663	0.1454	No	No	consistent
9	0.893	0.6858	0.2072	No	No	consistent
10	1	1	0	Yes	Yes	consistent

Table S10: A comprehensive analysis was performed to assess the significance of the association between P-values and frequently occurring genes in the GO (BP, CC, and MF), KEGG, WikiPathways, Reactome, BioCarta, BioPlanet, and DsigBD databases

Database	Term	P-value	Adjusted P-value	Genes
GO_Biological_Process	Regulation of Apoptotic Process (GO:0042981)	0.003868054783571444	0.09982545942863763	AHI1;DKKL1;PTMA;IGF1R
	Positive Regulation of Protein Localization to Chromosome, Telomeric Region (GO:1904816)	0.00474139160939957	0.09982545942863763	PML
	Cytochrome Metabolic Process (GO:1903604)	0.00474139160939957	0.09982545942863763	CYP2U1
	Regulation of Protein Localization to Chromosome, Telomeric Region (GO:1904814)	0.00474139160939957	0.09982545942863763	PML
	Regulation of Translational Initiation in Response to Stress	0.005687115738983334	0.09982545942863763	PML

	(GO:0043558)			
	Notch Receptor Processing (GO:0007220)	0.00663198906678106	0.09982545942863763	GALNT11
	Dendritic Spine Maintenance (GO:0097062)	0.00663198906678106	0.09982545942863763	IGF1R
	Negative Regulation of Vasoconstriction (GO:0045906)	0.00663198906678106	0.09982545942863763	DOCK4
	Positive Regulation of Vascular Associated Smooth Muscle Cell Migration (GO:1904754)	0.00757601220630419 25	0.09982545942863763	DOCK4
	Protein O-linked Glycosylation via Threonine (GO:0018243)	0.00851918580033512 8	0.09982545942863763	GALNT11
GO_Cellular_Component	Transferase Complex, Transferr	0.00757601220630419 25	0.11622188003447768	IGF1R

	ing Phosphorus-Containing Groups (GO:0061695)			
	Primary Lysosome (GO:0005766)	0.010402987018789436	0.11622188003447768	OLFM4
	Protein Kinase Complex (GO:1902911)	0.020703445636816073	0.11622188003447768	IGF1R
	Pigment Granule (GO:0048770)	0.022565311 42960272	0.11622188003447768	SYTL2
	P Granule (GO:0043186)	0.023494986 40716487	0.11622188003447768	TDRD9
	Exocytic Vesicle (GO:0070382)	0.027205316711375317	0.11622188003447768	SYTL2
	Secretory Vesicle (GO:0099503)	0.029055470 00861942	0.11622188003447768	SYTL2
	Melanosome (GO:0042470)	0.029055470 00861942	0.11622188003447768	SYTL2
	Nucleolus (GO:0005730)	0.038924811 18124631	0.12809792367 50563	DOCK4;PML;IGF1R
	Nuclear Lumen (GO:0031981)	0.040030601 14845509	0.12809792367 50563	DOCK4;PML;IGF1R

GO_Molecular_Function	Riboflavin Transmembrane Transporter Activity (GO:0032217)	0.00474139160939957	0.09611585348385629	SLC22A14
	Semaphorin Receptor Activity (GO:0017154)	0.009461510512395743	0.09611585348385629	PLXNA2
	Neurexin Family Protein Binding (GO:0042043)	0.009461510512395743	0.09611585348385629	SYTL2
	SUMO Binding (GO:0032183)	0.013222334171232074	0.09611585348385629	PML
	Insulin-Like Growth Factor Binding (GO:0005520)	0.014160424747060764	0.09611585348385629	IGF1R
	Receptor Antagonist Activity (GO:0048019)	0.016034072395465458385629	0.09611585348385629	DKKL1
	Ribosomal Small Subunit Binding (GO:0043024)	0.017904346732621168385629	0.09611585348385629	EIF4B
	Insulin Receptor Binding	0.017904346732621168385629	0.09611585348385629	IGF1R

	(GO:0005158)			
	Steroid Hydroxylase Activity (GO:0008395)	0.021634798076404317	0.09611585348385629	CYP2U1
	Notch Binding (GO:0005112)	0.021634798076404317	0.09611585348385629	GALNT11
KEGG	Pathways in cancer	1.6007464465940872e-16	1.3446270151390334e-14	HDAC1;STAT2;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	PI3K-Akt signaling pathway	5.8305603991696 7e-09	2.4488353676512615e-07	HGF;ERBB2;MDM2;JAK2;IGF1R;JAK1
	Epstein-Barr virus infection	2.4188230275102 53e-08	5.4686760603029 19e-07	HDAC1;STAT2;STAT3;MDM2;JAK1
	Proteoglycans in cancer	2.6041314572871046e-08	5.4686760603029 19e-07	HGF;ERBB2;STAT3;MDM2;IGF1R
	Signaling pathways regulating pluripotency of stem cells	5.0890052897207 45e-07	8.5495288867308 52e-06	STAT3;JAK2;IGF1R;JAK1
	Necroptosis	7.7815541271745 7e-07	8.8048359101190 95e-06	STAT2;STAT3;JAK2;JAK1
	JAK-STAT signaling pathway	8.3855580096372 34e-07	8.8048359101190 95e-06	STAT2;STAT3;JAK2;JAK1
	Hepatitis B	8.3855580096372 34e-07	8.8048359101190 95e-06	STAT2;STAT3;JAK2;JAK1
	Influenza A	1.0653381927458568e-06	9.9431564656279 95e-06	STAT2;JAK2;PML;JAK1

	Transcriptional misregulation in cancer	1.6522764653746114e-06	1.2881310503869374e-05	HDAC1;MDM2;PML;IGF1R
WiKipathway	Cancer Pathways WP5434	1.0038898366229864e-16	1.8070017059213757e-14	HDAC1;STAT2;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	EGFR Tyrosine Kinase Inhibitor Resistance WP4806	9.478968873949188e-13	8.5310719865542269e-11	HGF;ERBB2;STAT3;JAK2;IGF1R;JAK1
	IL6 Signaling WP364	3.8516300753255531e-09	2.3109780451953319e-07	HDAC1;STAT3;JAK2;JAK1
	Thyroid Stimulating Hormone TSH Signaling WP2032	1.3194101796692674e-08	5.9373458085117703e-07	STAT3;JAK2;IGF1R;JAK1
	IL10 Anti Inflammatory Signaling WP4495	1.9753256746527796e-08	7.1111724287500067e-07	STAT2;STAT3;JAK1
	ncRNAs Involved In STAT3 Signaling In Hepatocellular Carcinoma WP4337	2.5672603330014545e-08	7.7017809990004364e-07	STAT3;JAK2;JAK1
	Leptin Signalin	3.0122338236504474e-08	7.7457441179583362e-07	ERBB2;STAT3;JAK2;JAK1

	g WP2034			
	Prolactin Signaling WP2037	3.5645724228129814e-08	8.020287951329207e-07	ERBB2;STAT3;JAK2;JAK1
	Immune Response To Tuberculosis WP4197	1.379143686230442e-07	2.758287372460884e-06	STAT2;JAK2;JAK1
	Androgen Receptor Network In Prostate Cancer WP2263	1.7090436968772706e-07	2.992161265242917e-06	HGF;STAT3;MDM2;JAK1
Reactome	Disease	1.850219543728829e-10	5.08810374525428e-08	HDAC1;STAT2;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	Interleukin-20 Family Signaling	4.68480811645484e-10	6.441611160125407e-08	STAT2;STAT3;JAK2;JAK1
	Interleukin-27 Signaling	1.4818761302081364e-08	1.0187898395180938e-06	STAT3;JAK2;JAK1
	Interleukin-6 Signaling	1.4818761302081364e-08	1.0187898395180938e-06	STAT3;JAK2;JAK1
	Interleukin-35 Signalling	1.975325674652796e-08	1.086429121059038e-06	STAT3;JAK2;JAK1
	Diseases of Signal Transduction by Growth Factor Receptors and	2.474084645630725e-08	1.133955462580749e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2

	Second Messengers			
	Signaling by Receptor Tyrosine Kinases	6.905612889545374e-08	2.712919349464253 3e-06	HDAC1;HGF;ERBB2;STAT3;JAK2;IGF1R
	Signal Transduction	9.673158364802686e-08	3.325148187900923 4e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2;PML;IGF1R;JAK1
	Interferon Signaling	1.234792290092498e-07	3.456467839835088 7e-06	STAT2;STAT3;JAK2;PML;JAK1
	PIP3 Activates AKT Signaling	1.256897396 3036686e-07	3.456467839835088 7e-06	HDAC1;HGF;ERBB2;MDM2;PML
BioCarta	Role of ERBB2 in Signal Transduction and Oncology Homo sapiens her2Pathway	6.413108912119444e-10	3.014161188696138 3e-08	ERBB2;STAT3;JAK2;JAK1
	Stat3 Signaling Pathway Homo sapiens hstat3Pathway	5.033265573129969e-09	1.182817409685542 9e-07	STAT3;JAK2;JAK1
	IL 6 signaling pathway Homo sapiens hil6Pathway	2.567260333 0014545e-08	4.022041188368946e-07	STAT3;JAK2;JAK1
	IFN gamma	3.371314168 1840526e-06	3.961294147616262e-05	JAK2;JAK1

	signaling pathway Homo sapiens hifngPathway			
	IFN alpha signaling pathway Homo sapiens hifnaPathway	8.084763760961904e-06	7.599677935300419e-05	STAT2;JAK1
	IL22 Soluble Receptor Signaling Pathway Homo sapiens hil22bppathway	1.234519196297719e-05	9.670400370998e-05	STAT3;JAK1
	Sumoylation by RanBP2 Regulates Transcriptional Repression Homo sapiens hranbp2Pathway	2.040945030 25842e-05	0.0001370348 8060306536	HDAC1;MDM2
	EGF Signaling Pathway Homo sapiens hegfPathway	4.254530034649759e-05	0.0002499536395356733	STAT3;JAK1
	TPO Signaling	6.173695322587491e-05	0.0003224040 8906845783	STAT3;JAK2

	Pathway Homo sapiens hTPOPathway			
	PDGF Signaling Pathway Homo sapiens hpdgfPathway	7.2658832459491 73e-05	0.0003414965125596111	STAT3;JAK1
BioPlanet	Pathways in cancer	1.9517938047337523e-13	4.606233379171654e-11	HDAC1;HGF;ERBB2;STAT3;MDM2;PML;IGF1R;JAK1
	Interleukin-27-mediated signaling events	4.6848081164548 4e-10	5.528073577416713e-08	STAT2;STAT3;JAK2;JAK1
	Immune system signaling by interferons, interleukins, prolactin, and growth hormones	1.4298395540601624e-09	1.124807115860661 3e-07	STAT2;HGF;STAT3;JAK2;PML;JAK1
	Leptin signaling pathway	1.6217188052366822e-08	9.568140950896422e-07	ERBB2;STAT3;JAK2;JAK1
	Thyroid-stimulating hormone signaling pathway	2.2370327392347 91e-08	1.009789064313905 5e-06	STAT3;JAK2;IGF1R;JAK1
	Interleukin-22 soluble receptor	2.5672603330014545e-08	1.009789064313905 5e-06	STAT3;JAK2;JAK1

	signaling pathway			
	Interleukin-6 signaling pathway	3.012233823650474e-08	1.015553117687874e-06	HDAC1;STAT3;JAK2;JAK1
	Prolactin activation of MAPK signaling	3.7644908603965495e-08	1.1105248038169822e-06	ERBB2;STAT3;JAK2;JAK1
	Immune system	7.932256198009344e-08	2.049574689362683e-06	STAT2;HGF;STAT3;MDM2;JAK2;PML;JAK1
	STAT3 pathway	8.684638514248656e-08	2.049574689362683e-06	STAT3;JAK2;JAK1
DSigBD	curcumin CTD 00000663	9.624731064478704e-10	1.0510206322410747e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2;IGF1R
	Arsenenous acid CTD 00000922	1.125007154544032e-08	6.142539063810416e-06	HDAC1;HGF;ERBB2;STAT3;MDM2;JAK2;PML;JAK1
	Capsaicin CTD 00005570	2.6683516299472512e-08	9.712799933007994e-06	ERBB2;STAT3;MDM2;IGF1R;JAK1
	Vorinostat BOSS	5.6805663671864834e-08	1.0055255092706986e-05	HDAC1;ERBB2;STAT3;MDM2
	172889-27-9 BOSS	5.963163884052429e-08	1.0055255092706986e-05	ERBB2;STAT3;MDM2;JAK2
	NVP-TAE684 CTD 00004657	6.097651861537575e-08	1.0055255092706986e-05	HGF;ERBB2;STAT3
	resveratrol CTD 00002483	6.445676341478837e-08	1.0055255092706986e-05	HDAC1;HGF;ERBB2;STAT3;MDM2;PML;IGF1R;JAK1
	Erlotinib BOSS	1.11263735045227e-07	1.5187499833673485e-05	ERBB2;STAT3;JAK2;IGF1R
	podophyllotoxin	2.058149650923037e-07	2.497221576453285e-05	STAT3;MDM2;JAK2

	TTD 00010285			
	podophyllotoxin CTD 00006577	2.92922641574985e-07	3.159426968121176e-05	STAT3;MDM2;JAK2

Table S11. Receptor structures and docking grid parameters used for molecular docking of the anthrax hub proteins.

Gene	Organism	Structure source	Structure ID	Method	Resolution / pLDDT	Chains in receptor file	Grid center (x, y, z)
DKK4	Homo sapiens	Protein Data Bank	9X64	X-ray diffraction	1.83 Å	A, B	-7.297, -15.973, 2.254
FZD3	Homo sapiens	Protein Data Bank	8JHI	Cryo-EM	3.20 Å	A, B, C, N, R	121.880, 119.383, 98.337
FZD6	Homo sapiens	Protein Data Bank	8JHB	Cryo-EM	3.30 Å	A, B, C, N, R	119.359, 122.750, 136.538
FZD9	Homo sapiens	AlphaFold Protein Structure Database	AF-O00144-F1	Computational prediction	pLDDT 80.93	A	-7.085, 6.625, -10.132
WNT6	Homo sapiens	AlphaFold Protein Structure Database	AF-Q9Y6F9-F1	Computational prediction	pLDDT 83.01	A	[to be filled after re-docking]

Structures for DKK4, FZD3, and FZD6 were retrieved from the Protein Data Bank (PDB); structures for FZD9 and WNT6 are AlphaFold-predicted models (global pLDDT shown). The full deposited structure, including all co-crystallized chains, was used as the receptor. Grid centers are in Å and were defined as the geometric center of each receptor in AutoDockTools. All docking runs used a 40 × 40 × 40 Å search box, an exhaustiveness of 8, and an energy range of 4. PDB, Protein Data Bank; pLDDT, predicted local distance difference test.

Table S12: Complete molecular docking results for all protein–ligand pairs screened with AutoDock Vina.

Gene	Compound	PubChem CID	Best binding affinity, mode 1 (kcal/mol)	Number of poses reported
DKK4	Irinotecan	60838	-9.6	9
DKK4	Daunorubicin	30323	-8.3	9
DKK4	Celastrol	122724	-8.3	9

DKK4	Hesperidin	10621	-8.3	9
DKK4	Quercetin	5280343	-8	9
FZD3	Irinotecan	60838	-9.6	9
FZD3	Daunorubicin	30323	-7.8	9
FZD3	Celastrol	122724	-8.8	9
FZD3	Hesperidin	10621	-8.3	9
FZD3	Quercetin	5280343	-8.8	9
FZD6	Irinotecan	60838	-11.1	9
FZD6	Daunorubicin	30323	-8.3	9
FZD6	Celastrol	122724	-8.2	9
FZD6	Hesperidin	10621	-8.8	9
FZD6	Quercetin	5280343	-9.3	9
FZD9	Irinotecan	60838	-8.2	9
FZD9	Daunorubicin	30323	-7.6	9
FZD9	Celastrol	122724	-8.2	9
FZD9	Hesperidin	10621	-8.3	9
FZD9	Quercetin	5280343	-7.7	9
WNT6	Irinotecan	60838	-10	9
WNT6	Daunorubicin	30323	-7.7	9
WNT6	Celastrol	122724	-8	9
WNT6	Hesperidin	10621	-8.5	9
WNT6	Quercetin	5280343	-6.7	9

Binding affinity (kcal/mol) is the best-scoring pose (mode 1) reported by AutoDock Vina; more negative values indicate stronger predicted binding. Nine poses were reported per docking run. Compounds are identified by their PubChem compound identifier (CID). All runs used a 40 × 40 × 40 Å search box, an exhaustiveness of 8, and an energy range of 4. CID, compound identifier.

Supplementary Figures:

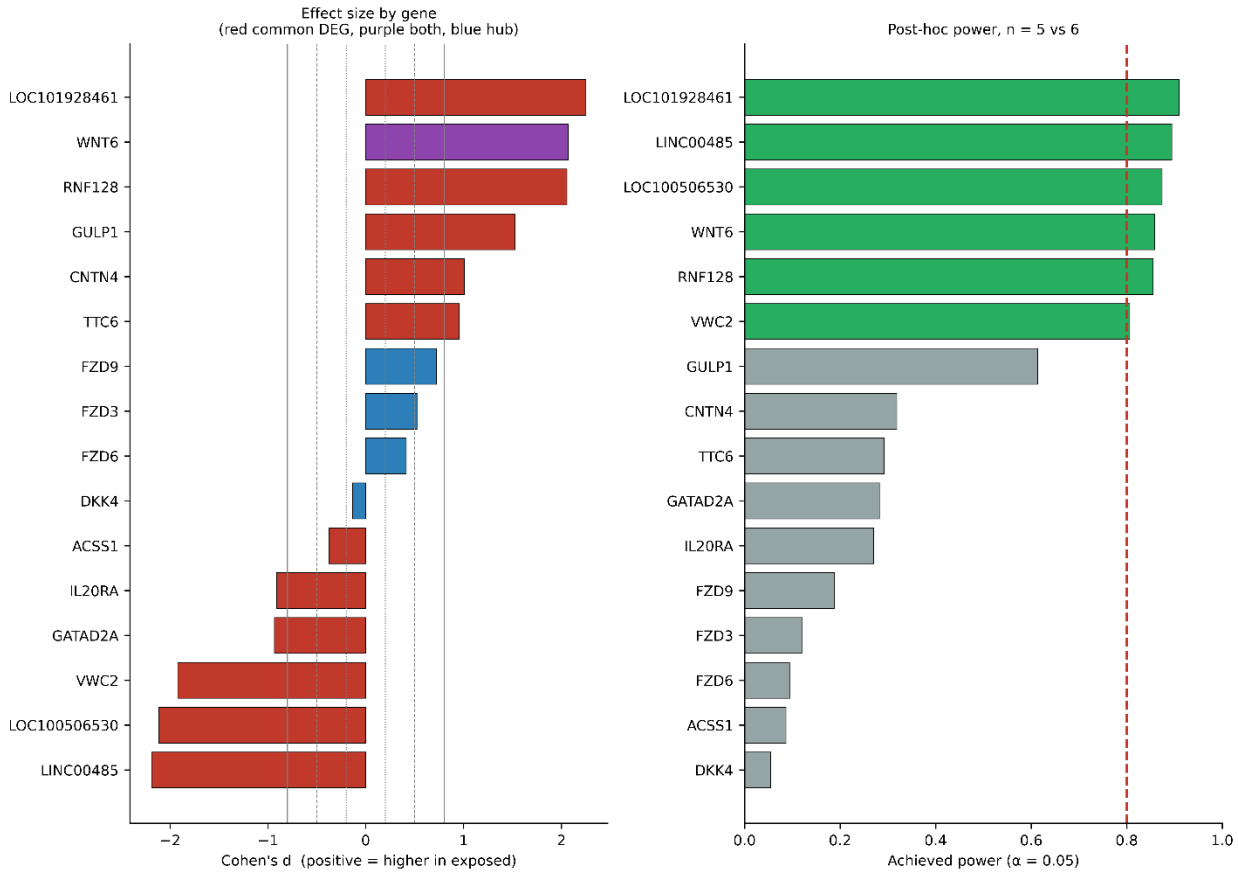


Figure S1: Effect size and achieved statistical power for the sixteen genes assessed. (A) Cohen's d, with reference lines at the conventional small, medium, and large thresholds. (B) Achieved power at a two-sided α of 0.05, with the 0.80 threshold marked.

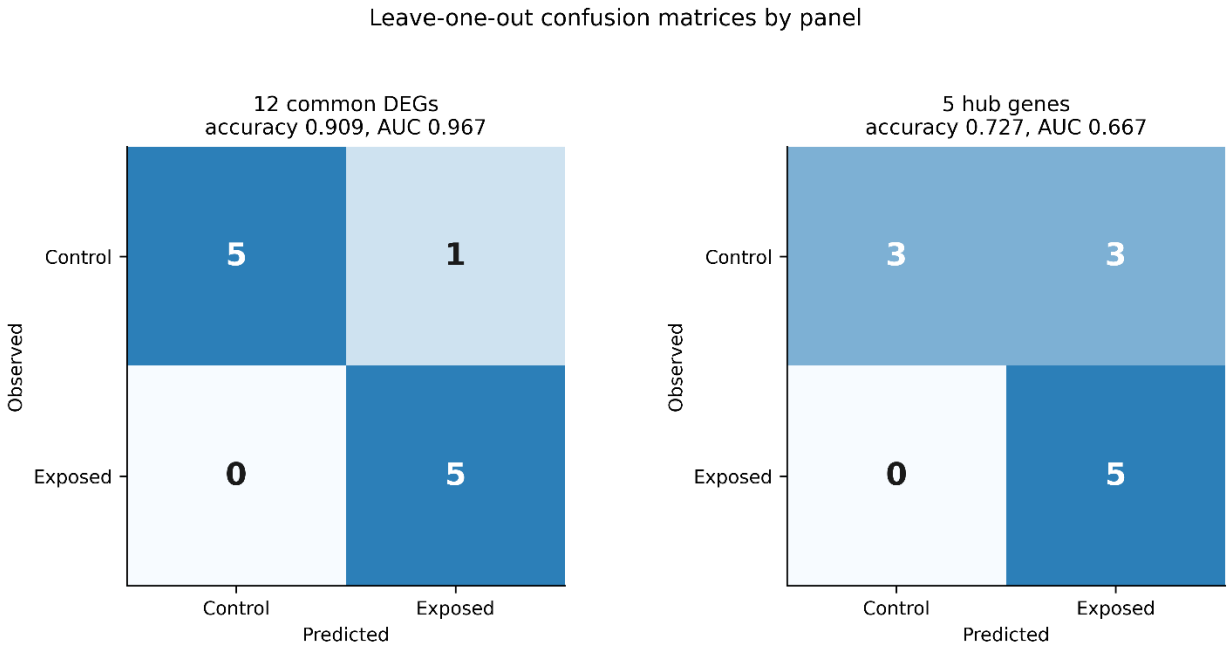


Figure S2: Leave-one-out confusion matrices for both panels in the combined discovery cohort (n = 11; 5 exposed, 6 control). Counts are cross-validated, each sample having been scored by a model refitted without it.

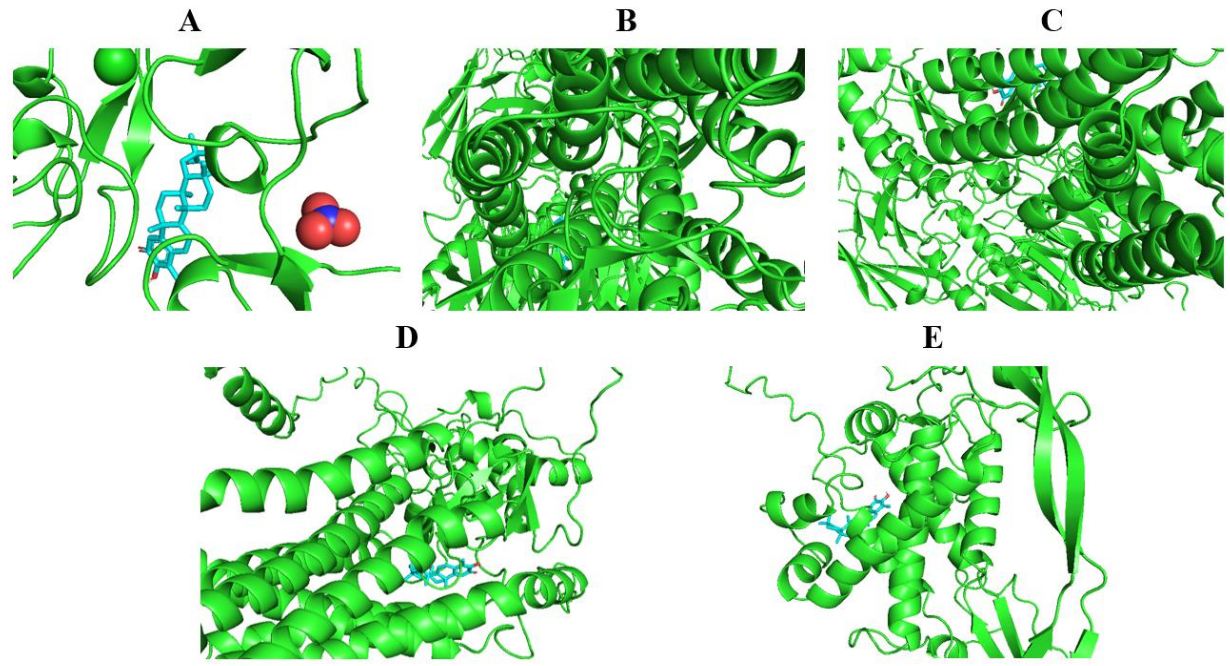


Figure S3: Docked poses of Celastrol in the five anthrax-associated hub proteins: (A) DKK4, (B) FZD3, (C) FZD6, (D) FZD9, (E) WNT6. Receptors are shown as ribbons, and the ligand is shown in stick

representation within the predicted binding site.

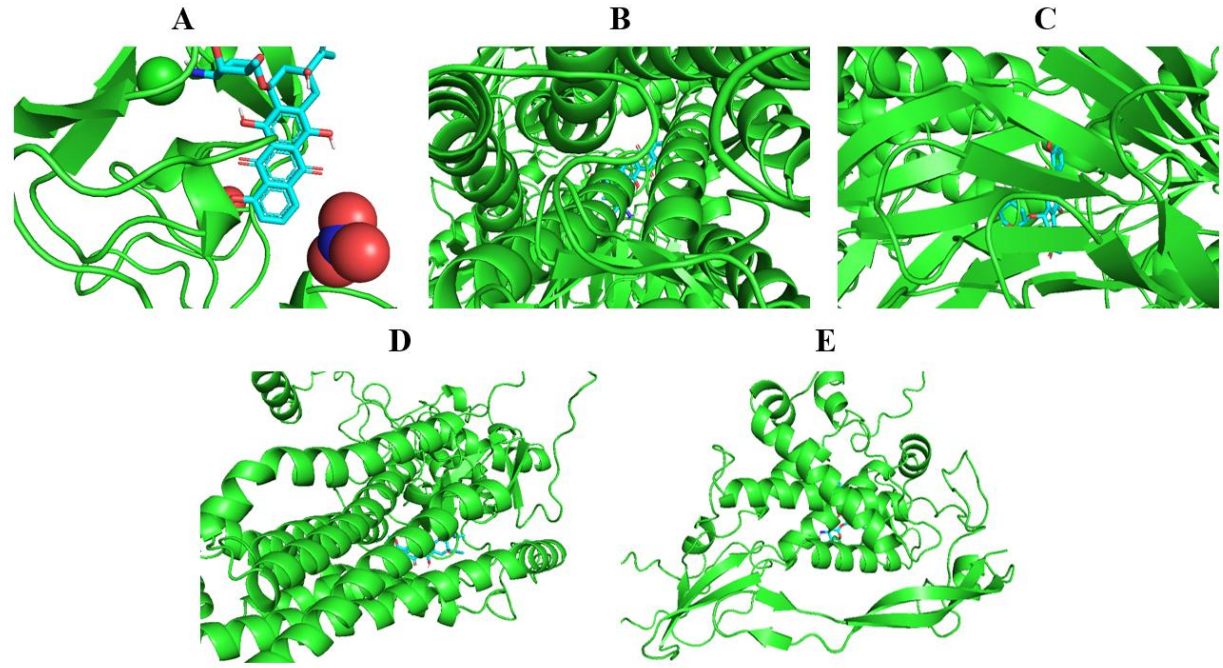


Figure S4: Docked poses of Daunorubicin in the five hub proteins: (A) DKK4, (B) FZD3, (C) FZD6, (D) FZD9, (E) WNT6.

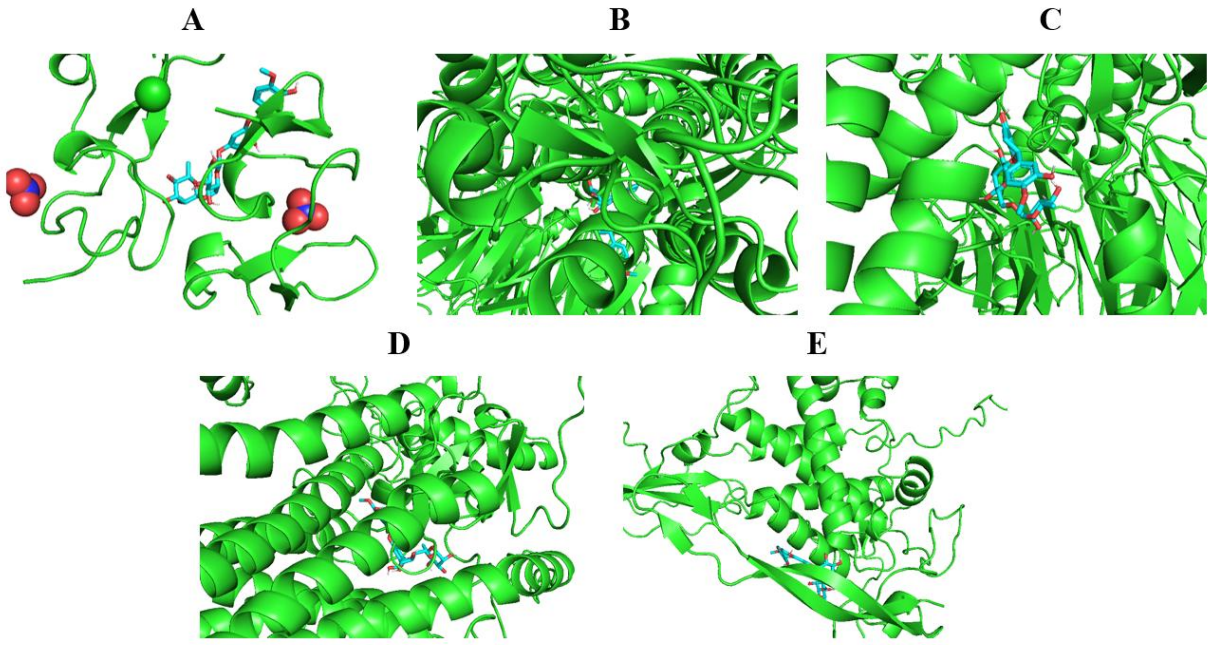


Figure S5: Docked poses of Hesperidin in the five hub proteins: (A) DKK4, (B) FZD3, (C) FZD6, (D) FZD9, (E) WNT6.

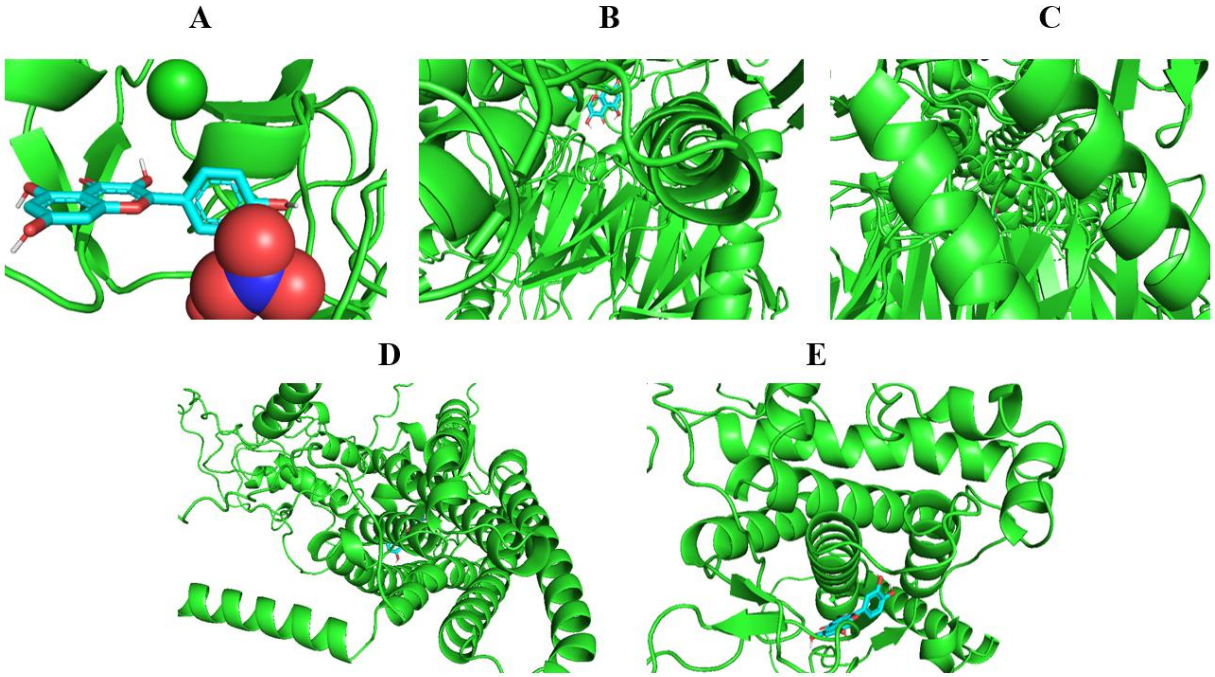


Figure S6: Docked poses of Quercetin in the five hub proteins: (A) DKK4, (B) FZD3, (C) FZD6, (D) FZD9, (E) WNT6.

Reproducibility

Analyses were performed in Python 3.13 (scikit-learn, statsmodels, SciPy, pandas) and R 4.6 (limma, pvclust, data.table, pROC, ggplot2). Networks were constructed in NetworkAnalyst and visualised in Cytoscape v3.10.3 with the CytoHubba plugin; docking used AutoDock Vina 1.2.0 with receptors prepared in AutoDockTools 1.5.7. A fixed random seed was applied throughout. Both series are available from the Gene Expression Omnibus under accessions GSE14390 and GSE12131.